STACKCALC SC-32 SCIENTIFIC RPN CALCULATOR

4-Register RPN Stack Engine, Dual Stand Modes & Hardware Guide

SERIES 1 • CARD 09 / 12 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES

CARD 09 / 12 • SC-32-HW

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

4-REGISTER RPN ENGINE & DUAL STANDS

Reverse Polish Notation (RPN) with 4 registers.
Numbers push onto stack; operators evaluate at once:
Stack Architecture: Regs T, Z, Y, X
Zero parentheses, zero ambiguity, zero errors.

CALIBRATED 32-KEYCAP TACTILE HARDWARE

- Display: 128×64 OLED with live 4-stack view
- Keycaps: 43 sculpted Shore 90A TPU keys over tactile switches
- Stack: Push on [ENTER], pop on op, LastX saved

OPERATING MODES: 11.6° DESK INCLINE & 78° PU



DESK STAND (11.6°)
Seated typing • Glare-free OLED

PUZZLE STAND (78°)
3-pc easel • Knocks flat to 6mm

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC	COLOR
SC-32 Calculator	1x	128×64 OLED	Obsidian
TPU Bumper Sleeve	1x	Shore 95A TPU	Black TPU
3-Piece Stand	1x	78° Easel	Matte Charcoal
Desk Stand Caddy	1x	11.6° Incline	Matte Charcoal
USB-C Cable	1x	1.0m Braided	Stealth Black
Mailer Box	1x	6×4 In Case	Kraft Box

TOTAL: Calculator + TPU Sleeve + Stands + Cable + Box = 6 Parts

POWER CONTROL & ANNUNCIATORS

- Instant Power Off: Tap [f] then [ON/OFF] to shut down.
- Auto-Sleep Standby: Sleeps after 5 min idle (<15 μA).
- Non-Volatile Memory: Registers (X, Y, Z, T) persist.
- Annunciators: RAD/DEG, [f], [g], battery on OLED header.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Press [f] [OFF] to turn off, or let auto-sleep preserve your stack. Dock in the Desk Stand or slip into the TPU bumper sleeve and drop in your backpack with zero loose parts!

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Any algebraic expression can be evaluated unambiguously without parentheses using a postfix stack.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: 4-REGISTER RPN STACK & EXPRESSION TREES

Step-by-Step Hands-On Tutorial: Pythagorean Hypotenuse $\sqrt{(3^2 + 4^2)} = 5$

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CARD 09 / 12 • SC-32-HW

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/sc32-calculator/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS MP5 (Use Tools St...)
- CCSS MP6 (Attend to Precision)
- TEKS 111.39.c.1.C (Manipulative...)

Grades 8–16 STEM, Physics & Engineering

GUIDING INQUIRY

Where does the first number wait safely on the stack while you are entering the second number?

① STEP 1: ENTER 3 AND 4 ONTO STACK

Power on the SC-32. Key: 3 [ENTER] 4 [ENTER].
Stack state: Register Y holds 3.0000; Register X holds 4.0000.
In RPN, numbers always enter before operators!

② STEP 2: SQUARE BOTH REGISTERS

Press [⁻¹] [x²] on 4: Register X becomes 16.0000.
Press swap [x⇄y]: Register X becomes 3.0000; Y holds 16.0000.
Press [⁻¹] [x²] on 3: Register X becomes 9.0000.

③-④ STEPS 3 & 4: COLLAPSE STACK & SQRT

3. ADD SQUARES: Press [+]. The stack drops: 16.0000 + 9.0000 = 25.0000 in X!
4. SQUARE ROOT: Press [√x]. Result: 5.0000!
Combined Keystrokes: 3 [⁻¹] [x²] 4 [⁻¹] [x²] [+] [√x] → 5.

⑤ CORE PRINCIPLE: THE FREEDOM FROM PARENTHESES

Evaluating $\sqrt{(3^2 + 4^2)}$ in traditional algebraic mode requires nested parentheses. In RPN, you calculate from the inside out, exactly as you think! Intermediate numbers sit patiently on the stack until an operator consumes them. Zero parentheses, zero syntax errors!

★ EXTENSION INVESTIGATION: KINETIC ENERGY: $E = \frac{1}{2} m v^2$

Calculate kinetic energy for mass m=12 and velocity v=5: 0.5 [ENTER] 12 [×] 5 [⁻¹] [x²] [×]
→ 150.0000! Zero parentheses! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL
stackcalc.io/parts

STACKCALC SC-32 HARDWARE, POWER & DUAL STANDS

DESK INCLINE (11.6°) • PUZZLE EASEL (78°)

1. POWER & AUTO-SLEEP

INSTANT SHUTDOWN: [f] [OFF]
Electronic calculator with zero loose blocks to pack. Tap [f] then [ON/OFF] to shut down immediately.

ULTRA-LOW AUTO-SLEEP
Standby engages automatically after 5 minutes idle (<15 μA draw). Battery lasts weeks on standby.

NON-VOLATILE FLASH RAM
Registers (X, Y, Z, T), LastX, angular modes, and memories persist completely across power cycles.

INSTANT RESUME ON KEY
Press [ON] or tap any key to wake immediately with previous calculation waiting where you left it.

ZERO PACK-UP OVERHEAD
Turn it off, slip into TPU sleeve, and go.

2. DESK STAND (11.6°)

1.0 IN REAR ELEVATION
Elevates rear by 25.4 mm, giving an optimal 11.6° forward slope for seated one-handed typing.

ANTI-GLARE OLED VIEW
Angled viewing geometry eliminates ceiling fluorescent light reflections across the OLED panel.

INTEGRATED CABLE TROUGH
Rear pass-through connects braided USB-C cable for continuous desktop charging during operation.

ROCK-SOLID FOOTPRINT
Wide stance chassis with non-slip base prevents sliding or tipping during rapid tactile key entries.

ACTIVE OPERATION MODE
Keep docked on desk alongside your workspace.

3. PUZZLE STAND (78°)

3-PIECE FRICTION DOVETAIL
Tool-free interlocking trusses snap together by hand in 5 seconds with precision 0.20 mm slide tolerance.

UPRIGHT PORTRAIT EASEL
Docks calculator at 78° upright portrait viewing beside open STEM textbooks and lab worksheets.

DUAL INTEGRATED BAYS
Front cradle holds calculator securely; rear bay docks TPU travel sleeve or reference card upright.

KNOCKS FLAT TO 6.0 MM
Disassembles in 2 seconds into 3 slim plates that slip flat into a pencil case or backpack pouch.

LAB & SHELF DISPLAY
Optimal reading angle while hands are full.

HARDWARE WORKFLOW & TRAVEL SETUP

1. SEATED DESK CALCULATING (DESK STAND):
Set travel caddy on desk and seat calculator onto 11.6° incline. Connect USB-C for continuous power during long study sessions.

2. HANDS-ON LAB INVESTIGATIONS (PUZZLE STAND):
Slide the two puzzle stand trusses into cradle shelf until flush. Dock calculator at 78° to read values while building physical models.

3. POWER DOWN & SLEEP:
When finished, press [f] [OFF] to shut down, or let auto-sleep preserve your stack. There are no loose parts to count or pack!

4. TRAVEL PROTECTION & KNOCKDOWN:
Slip calculator into Shore 95A TPU bumper. Knock puzzle stand flat to 6mm. Pack both into travel caddy for drop-proof transit.

SC-32-HW: HARDWARE SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC HARDWARE AUDIT

[✓] OLED DISPLAY & POLARIZER AUDIT:
128×64 high-contrast display verified for zero pixel defects, uniform brightness, and 4-line stack viewing.

[✓] 43 TACTILE SWITCHES & TPU KEYS:
Actuation force verified at 1.2 N ± 0.1 N with positive tactile feedback across all 43 keys.

[✓] PUZZLE STAND DOVETAIL SLIDE FIT:
Interlocking trusses seat flush with 0.20 mm precision friction fit for wobble-free 78° docking.

[✓] USB-C POWER DELIVERY & SLEEP CURRENT:
LiPo charging controller and ultra-low standby current (<15 μA) verified across power-down cycles.

[✓] SHORE 95A FLEXIBLE TPU MEMBRANE:
Perimeter shock-absorbing bumper cushion molded to exact 0.25 mm clearance for drop protection.

[✓] STACK STATE MEMORY RETENTION:
Registers X, Y, Z, T, LastX, and angular modes tested for 100% data integrity through simulated auto-sleep.

RPN HARDWARE COMPATIBILITY

- Full scientific transcendental engine (sin, cos, tan, ln, e^x, y^x)
- Statistical registers ($\Sigma+$, $\Sigma-$, mean, standard deviation)
- Postfix stack eliminates operator precedence ambiguity
- Digital twin of physical manipulative heights across all kits!

TOTAL HARDWARE INVENTORY: 6 PIECES

Verified by Electronics Tech & Hardware Test Bench.

Inspection Pass • Mountain View, CA

LIFETIME HARDWARE GUARANTEE

NO COMPONENT LEFT BEHIND™

Hardware components wear or get misplaced. We've got you covered. Replacement puzzle stand trusses (piece_stand_front, piece_stand_rear), TPU protective sleeves, and braided USB-C cables are permanently stocked at stackcalc.io/parts with same-day dispatch ($1.99–$7.50).
PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D HARDWARE INTERACTIVE DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of RPN stack dynamics
- Interactive 3D explode view of SC-32 chassis & stand
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California